

Ararat Tonoyan

SENIOR BLOCKCHAIN ENGINEER · WEB3 ADVISOR

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Summary

Senior blockchain engineer with 11 years in software and 9 years in EVM. Top 27 Solidity Developer globally on Upwork in 2020. Shipped 100+ smart contracts across DeFi, tokenization, and RWA for 60+ clients in Europe and North America. Currently advising the Central Bank of Armenia and state institutions on Web3 regulation through Web3 Armenia Foundation. Open to remote and relocation.

Experience

Evangelist

Yerevan, Armenia

Web3 Armenia Foundation

Apr 2023 - Present

- Advise the Central Bank of Armenia on international crypto regulatory practices and frameworks
- Train the State Revenue Committee on crypto taxation methodology and on-chain reporting
- Educate Armenian Police on on-chain forensics, tracing tools, and investigation workflows
- Support grant diversification and ecosystem fundraising for early-stage Web3 projects in Armenia

Solidity Engineer

Yerevan, Armenia

40 Acres Finance

Apr 2025 - Feb 2026

- Developed ve(3,3) tokenomics contracts for a DeFi lending protocol
- Implemented self-repaying loan logic on top of ERC-4626 vaults, where yield from underlying collateral progressively settles borrower debt without active repayment
- Built frontend integration for protocol interactions using viem and wagmi

Independent Research & Consulting

Yerevan, Armenia

Self-directed

Oct 2021 - Mar 2025

- Selective advisory engagements with Web3 founders and small teams on protocol architecture
- Independent research across post-Merge Ethereum, Account Abstraction (ERC-4337, EIP-7702), L2 design, ve(3,3) systems, and ERC-3643 / RWA frameworks — directly informing later production work
- Authored long-form technical content on smart contract security, gas optimization, and the historical evolution of programmable money

Senior Blockchain Engineer

Yerevan, Armenia

Upwork (Top Rated, 100% Job Success Score)

Jan 2018 - Oct 2021

- Delivered 100+ smart contracts and decentralized protocols across DeFi, tokenization, and enterprise blockchain for 60+ clients in Europe and North America. Owned full lifecycle on each engagement
- Translated complex protocol mechanics for non-technical stakeholders, producing decisions that held up under real-world conditions months after delivery
- Maintained Top Rated status and 100% positive feedback across all engagements with no security incidents reported on deployed contracts

Lead Instructor — Solidity Program

Moscow, Russia

OTUS Online Education

Jun 2023 - Sep 2023

- Designed and delivered a Solidity re-education program for working developers transitioning into smart contract engineering
- Authored full curriculum covering contract architecture, security patterns, gas optimization, and production deployment scenarios
- Hired and coordinated three instructors, kept cohort size at 15 to preserve technical depth in feedback and discussion

Web & Blockchain Developer

Melbourne, Australia

Virtually Human (ZED.RUN)

Jan 2020 - Oct 2020

- Cut per-transaction cost from \$10 to \$0.01 by integrating Biconomy gasless flows and migrating core transaction logic to Polygon, directly unblocking user growth
- Implemented permit-based message signing to remove friction from on-chain user onboarding
- Designed the microservice architecture supporting the NFT gaming backend

Smart Contract Lead

Kyiv, Ukraine

First Bridge

May 2018 - Jan 2020

- Sole blockchain engineer across 10+ client engagements with corporates of varying maturity across multiple countries
- Owned full technical scope per project: architecture, development, third-party integrations
- Delivered a senior blockchain function to clients without requiring them to build internal teams

Smart Contract Lead

Moscow, Russia

Jury Online

Jan 2019 - Apr 2019

- Architected the smart contract system for a \$10M ICO, including a RICO-style structure and an on-chain arbitration protocol resolving disputes through token-weighted voting — predating mainstream on-chain governance
- Built the REST API layer connecting blockchain backend to frontend and led UI integration across the team

Smart Contract Developer

Saint Petersburg, Russia

QuantNet

Jan 2019 - Sep 2019

- Built a Merkle tree-based Proof of Data Existence (PoDE) verification system for transparent integrity checks without exposing underlying content
- Authored and open-sourced a Python microservice for off-chain verification, applicable to compliance and audit scenarios requiring proof of data state at a point in time

Web & Smart Contract Developer

New Jersey, USA

Leios

Jun 2018 - Oct 2019

- Worked on a fintech startup tackling international money transfers and bank integrations. Handled architectural planning for the blockchain layer, built the product from scratch through deployment, and developed a mobile MVP alongside the core smart contracts.

Frontend Developer

Yerevan, Armenia

BetConstruct

Nov 2016 - Dec 2017

- Built custom animations, UI components, and the lobby interface for a casino product serving 500+ daily active users
- Migrated key frontend modules to Angular.js, resolving long-standing performance bottlenecks

Frontend Developer

Voronezh, Russia

Doctornet.pro

Jun 2016 - Nov 2016

- Built Material Design dashboards and custom UI components tailored to individual client workflows
- Ran internal seminars for the development team focused on engineering practices and productivity

Frontend Developer (Internship)

Yerevan, Armenia

STDev

Dec 2014 - May 2016

- Built UI interfaces and implemented client-side business logic on production web applications
- Worked within an engineering team using pair programming as the primary method for code review and skill development

Projects

ZED.RUN — Gas-Optimized NFT Racing Platform

Biconomy Case Study

Web & Blockchain Developer (Virtually Human)

2020

- Reduced per-transaction cost from \$10 to \$0.01 by integrating Biconomy gasless flows and migrating core transaction logic to Polygon
- Removed wallet-popup friction with permit-based meta-transactions, directly unblocking user onboarding at scale
- Designed the microservice architecture supporting the NFT gaming backend

OTUS Solidity Re-Education Program

Habr · 27 Gas Optimizations

Lead Instructor & Curriculum Author

2023

- Authored full curriculum and led live sessions for a Solidity program targeting working developers transitioning into smart contract engineering
- Published three companion technical articles on Habr: a practical reference of 27 EVM-level gas optimizations, 18 unconventional Solidity tools, and the history of smart contracts from Nick Szabo to today
- Hired and coordinated three instructors, kept cohort size at 15 to preserve technical depth in feedback

QuantNet — Proof of Data Existence (PoDE)

Saint Petersburg, Russia

Blockchain Developer

2019

- Built a Merkle tree-based verification system providing transparent integrity proofs without exposing underlying data
- Authored and open-sourced a Python microservice for off-chain verification, applicable to compliance and audit workflows requiring proof of data state at a point in time

Skills

Smart Contracts & EVM	Solidity, Foundry, OpenZeppelin v5 , Hardhat, EVM internals, Gas optimization, Security patterns, Upgradeable proxies
DeFi & Standards	ERC-20 / 721 / 1155, ERC-4626 , ERC-4337 (Account Abstraction), EIP-7702, ERC-3643 (T-Rex / RWA), Permit2, Uniswap v3/v4, AMM design, ve(3,3) tokenomics
L2 & Cross-Chain	Base, Optimism, Arbitrum , Polygon, Chainlink CCIP, LayerZero, Bridge architecture
Web3 Tooling	viem, ethers.js v6 , wagmi, RainbowKit, The Graph, Tenderly, Etherscan / Blockscout
Frontend	TypeScript, React , Next.js, Vue / Nuxt, TailwindCSS
Backend & Infrastructure	Node.js , Python, PostgreSQL, Redis, Microservices, Cloud functions, Docker
Security & Research	Smart contract auditing , Threat modeling, Static analysis (Slither), Fuzzing (Echidna, Foundry), On-chain forensics
Spoken Languages	English (Fluent) , Russian (Native), Armenian (Native)

Education

National Polytechnic University of Armenia

Yerevan, Armenia

B.S. in Robotics Technology

September 2012 - June 2016

- Industrial robotics planning and design of automated assembly systems
- Design of automated systems for production efficiency and precision in industrial environments
- Foundation in systems thinking and engineering discipline that shaped a later focus on smart contract architecture